INSTITUTION
Taita Taveta National Polytechnic

DEPARTMENT
Mechatronics Engineering

FINAL YEAR PROJECT

Smart Hospital Queue Management System

with Triage and SMS Notification

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Introduction

Current Challenge: Long Queues

Healthcare facilities often experience long patient queues, which leads to significant delays and inefficiency in service delivery.

The Manual System Flaw

Many hospitals still rely on manual queue systems where patients must wait physically without knowing when they will be served.

Technological Solution

Advancements in embedded systems allow automation of such processes to improve efficiency and patient satisfaction.

Project Proposal

This project introduces a Smart Hospital Queue Management System that

Problem

Physical waiting, unknown wait times, and inefficiency.

Solution

Automated registration, triage prioritization, and SMS notifications.

Problem Statement

Traditional hospital queue systems present several operational challenges:

Long Waiting Times
Patients endure excessive and unpredictable wait times in physical queues.

Poor Communication
Lack of real-time updates creates a disconnect between patients and staff.

Patients Missing Turns
Physical absence often results in skipped service and confusion.

Lack of Prioritization
Manual systems often fail to effectively identify and prioritize emergency cases.

Inefficient Patient Flow
Disorganized management leads to crowded waiting areas and bottlenecks.

Time Wasted
Productive hours lost due to physical queuing requirements.

Staff Overload
Manual management diverts medical staff from core healthcare duties.

Critical Risks
High risk of delayed treatment for urgent triage cases.

Project Objectives

MAIN OBJECTIVE

To design and implement a **smart hospital queue management system** that improves patient flow and communication using **embedded technology**.

SPECIFIC OBJECTIVES

1. Implement patient **self-registration** using a keypad
2. Introduce **triage-based prioritization**
3. Generate **automatic token numbers**
4. Display queue information on an **LCD screen**
5. Send **SMS notifications** to patients regarding their queue status

System Components

The proposed system integrates the following hardware components to manage the queue efficiently.

Arduino Uno

Acts as the **Main System Controller**, processing all inputs from the keypad and managing outputs to the display and GSM module.

3×4 Matrix Keypad

Serves as the **Patient Input Interface** allowing users to select triage levels and enter their phone numbers.

16×2 LCD Display

Functions as the **System Output Display**, showing real-time token numbers, queue status, and instructions.

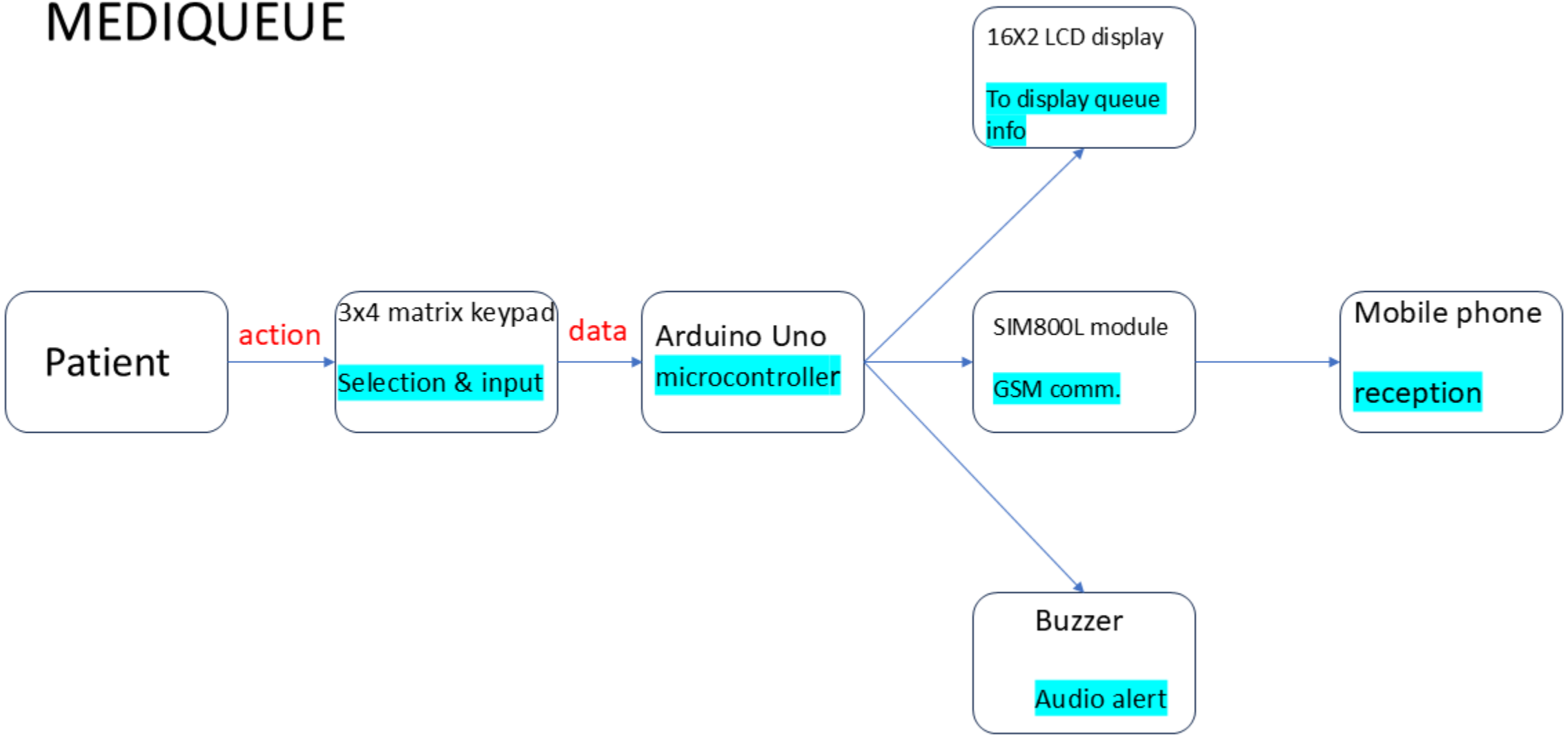
SIM800L GSM Module

Handles **SMS Communication**, sending token confirmations and queue alerts directly to patients' phones.

Buzzer

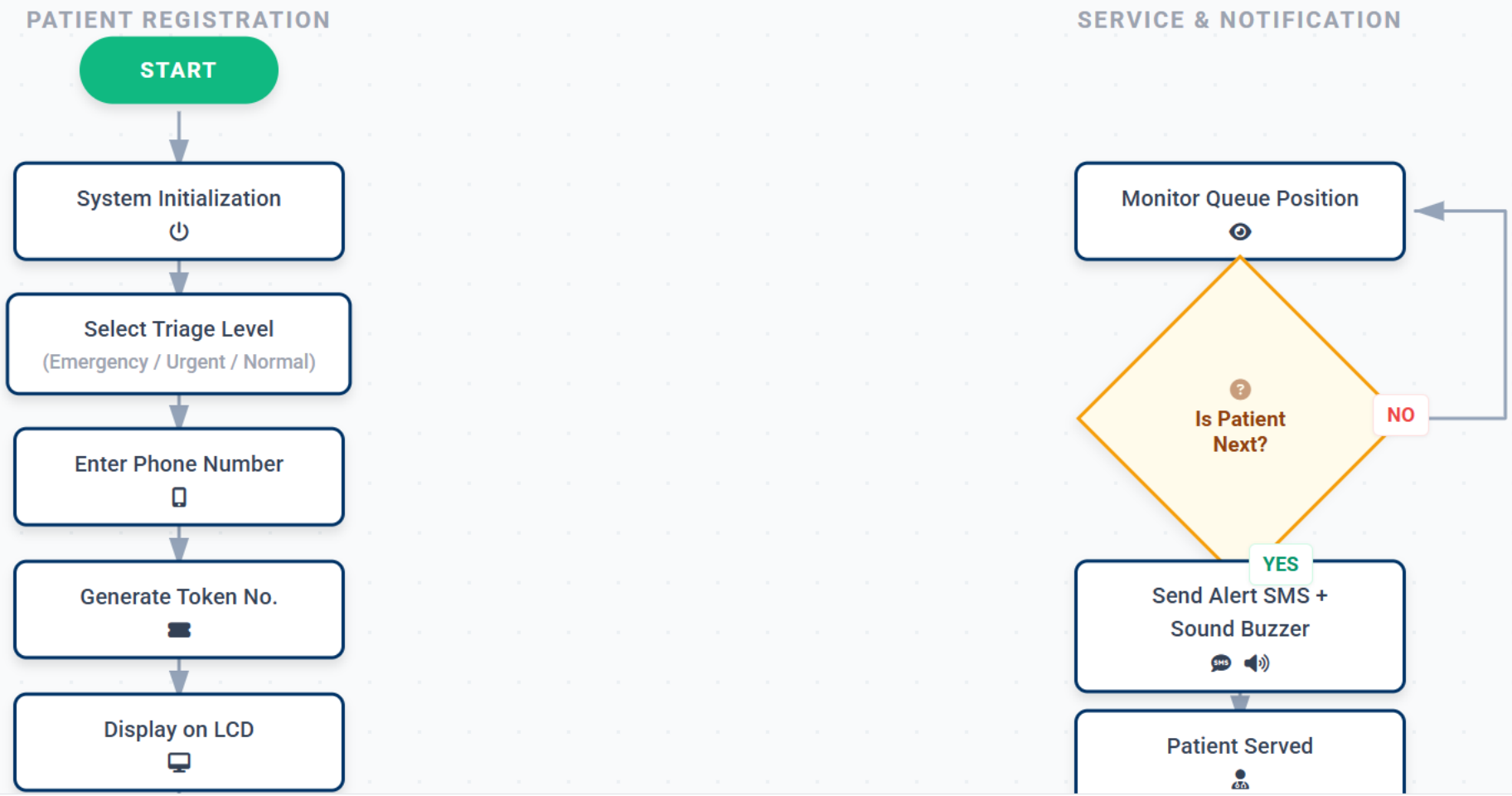
Provides **Audio Notification** to alert waiting patients when it is their turn to be served.

PROCESS FLOW DIAGRAM FOR MEDIQUEUE



System Flowchart

Start/End | Process | Decision



System Operation

The automated workflow ensures seamless patient management from arrival to service.

1

Triage Selection

Patient selects triage category (Emergency, Urgent, Normal) via keypad.

2

Phone Input

Patient enters their mobile phone number for notifications.

3

Token Generation

System processes input and generates a unique token number.

4

LCD Display

Token number and queue details are displayed on the screen.

5

SMS Confirmation

Token details sent to patient via SMS immediately.

6

Queue Alert

Alert SMS sent when patient is next in line.

7

Service Turn

Buzzer sounds and final SMS sent for service.

Advantages of the System

Reduces Uncertainty

Patients receive real-time updates on their queue status, eliminating the anxiety of unknown waiting times.

Improves Workflow Efficiency

Automates the flow of patients, reducing congestion in waiting areas and optimizing staff utilization.

Intelligent Triage Prioritization

Ensures emergency and urgent cases are automatically identified and prioritized over normal checkups.

Enhanced Communication

Direct SMS notifications keep patients informed even if they step away from the waiting area.

Cost-Effective Implementation

Time Saving
Optimized patient flow

Better Care
Critical cases first

Smart Tech
Low maintenance & automated

✓ System Efficiency Increased

Limitations of the System

Data Storage

The system currently has limited data storage capacity due to the microcontroller's internal memory constraints.

Database Integration

It operates as a standalone unit and does not currently integrate with existing hospital information databases.

Service Point Scale

The design supports only a single service point, requiring replication for multi-department use.

Network Dependency

Requires consistent GSM network availability; SMS notifications will fail in areas with poor signal coverage.

Current Constraints

Embedded hardware limitations restrict large-scale data handling and connectivity.

Future Outlook

These limitations can be effectively addressed in future system upgrades and cloud integration.

Future Improvements

Mobile Application

Develop a dedicated app allowing patients to monitor queue status remotely and receive real-time updates.

HIS Integration

Seamless integration with existing Hospital Information Systems (HIS) to sync patient data automatically.

Cloud Database

Transition to cloud-based storage for centralized, scalable, and secure patient records management.

Multi-Doctor Management

Expand system capability to manage multiple queues simultaneously for different doctors or departments.

Online Scheduling

Implement an online appointment booking system to reduce walk-in congestion and

Project Impact

Scalability
Expansion to cloud and multi-department support.

Usability
Enhanced user experience via mobile access.

Conclusion

Automated Solution

The Smart Hospital Queue Management System provides an effective automated solution for managing patient queues efficiently, successfully replacing traditional manual methods.

Enhanced Workflow & Experience

By integrating patient input via keypads, automatic token generation, and SMS notifications, the system significantly improves hospital workflow and enhances the overall patient experience.

Real-World Application

The system demonstrates how embedded technology can be practically applied to solve real-world healthcare challenges, bridging the gap between engineering theory and medical service needs.

Project Delivery

Successful implementation of hardware and software components for queue automation.

Healthcare Impact

Reduced waiting times, better communication, and improved service delivery.